

ELBaph: AWS Load Balancer Attack-Path Analysis

ELBaph: AWS Load Balancer Attack-Path Analysis - Francesco Lacerenza, Mohamed Ouad, Doyensec.

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README.md

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ELBaph - AWS Elastic Load Balancer Configuration Auditor

[[image: Doyensec Research Island]](<https://doyensec.com/research.html>) [image: GitHub]

A whitebox CLI tool for pentesters and security engineers to enumerate AWS Application Load Balancers and detect common routing misconfigurations that expose internal services to the public internet.

Built by [Doyensec](#) as part of **CloudSec Tidbits Season 2**.

Check reference (behaviour, samples): docs/elbaph-checks-documentation.md.

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Purpose

ELBaph is an active/passive auditing framework written in Go. Its goal is to identify routing misconfigurations and logical vulnerabilities across AWS Elastic Load Balancers (ELBs) that typical security posture management tools (CSPMs) miss.

Instead of just checking if logging is enabled or if a WAF is attached, ELBaph:

- Evaluates listener conditions (paths, host headers, source IPs).
- Cross-references Target Groups and their EC2 instance members across different ALBs.
- Analyzes interactions with surrounding network components (VPCs, CloudFront distributions).
- Performs targeted HTTP reachability probes to validate if backend services are truly exposed.

Installation

```
git clone https://github.com/doyensec/elbaph.git
cd elbaph
go build -o elbaph .
```

Requires Go 1.24 or later.

Usage

Scan a region - findings printed live, output folder created automatically

```
elbaph scan --region us-east-1
```

Scan multiple regions using an AWS profile

```
elbaph scan --all-regions -p my-pentest-profile
```

Inline access key + secret (optional session token for STS creds)

```
elbaph scan -r us-east-1 --aws-access-key-id AKIAEXAMPLE --aws-secret-access-key '...'
```

Shared credentials file (INI, same layout as ~/.aws/credentials) and profile name

```
elbaph scan -r us-east-1 --aws-credentials-file ./readonly.creds -p customer-audit
```

Route active HTTP probes through a remote proxy.

Useful when running the tool from an internal network or corporate VPN

to mimic an external vantage point and avoid false-positive reachability results.

```
elbaph scan -r us-east-1 --proxy http://remote.vantage.point:8080
```

Just output everything to a generic "out" folder as JSON and disable terminal colors

```
elbaph scan -r eu-west-1 -d out -o json --no-color
```

Run only specific checks

```
elbaph scan --region us-east-1 --checks alb-external-reachability,alb-ip-gate-bypass
```

Also export a structured report (alongside the always-created text folder)

```
elbaph scan --region us-east-1 --output json
```

```
elbaph scan --region us-east-1 --output markdown
```

```
elbaph scan --region us-east-1 --output sarif
```

Write the structured export to a specific path

```
elbaph scan --region us-east-1 --output sarif --file /tmp/results.sarif
```

Suppress terminal colours (useful in CI)

```
elbaph scan --region us-east-1 --no-color
```

Flags

Flag	Default	Description
-r, --region	us-east-1	AWS region to scan (ignored as sole region when --all-regions is set)
--all-regions	-	Scan every region enabled for the account (ec2:DescribeRegions)
-p, --profile	-	AWS named profile (with a custom credentials file, selects the profile block; otherwise uses shared config as usual)

Flag	Default	Description
<code>--aws-access-key-id</code>	-	Access key; must be used with <code>--aws-secret-access-key</code> ; mutually exclusive with <code>--aws-credentials-file</code>
<code>--aws-secret-access-key</code>	-	Secret key; must be used with <code>--aws-access-key-id</code>
<code>--aws-session-token</code>	-	Optional session token when using temporary (STS) inline credentials
<code>--aws-credentials-file</code>	-	Path to a credentials INI file (same format as <code>~/.aws/credentials</code>); mutually exclusive with the inline key flags; combine with <code>-p</code> for a non-default profile
<code>-c, --checks</code>	all	Comma-separated list of check IDs to run
<code>-d, --dir</code>	elbaph- <timestamp>	Output directory name
<code>-o, --output</code>	-	Also export: <code>json \</code>
<code>-f, --file</code>	<dir>/report. <ext>	Path for the structured export
<code>--proxy</code>	-	Optional HTTP proxy URL for active probing checks
<code>--vhost-probe-hosts-file</code>	-	For <code>alb-vhost-probe</code> : newline-separated hostnames merged after listener rules and Route 53 alias names but before the Route 53 zone corpus, so your list is not cut off when the corpus hits the candidate cap (Source: file-corpus)
<code>--vhost-probe-path</code>	/	For <code>alb-vhost-probe</code> : URL path for every GET probe (must start with <code>/</code> , e.g. <code>/api/health</code>)
<code>--no-color</code>	-	Disable terminal colour

AWS credentials

By default ELBaph uses the **AWS SDK default credential chain** for the scan region (and `us-east-1` for global calls such as CloudFront and `DescribeRegions` when `--all-regions` is set): environment variables (`AWS_ACCESS_KEY_ID` , `AWS_SECRET_ACCESS_KEY` , `AWS_SESSION_TOKEN`), shared files under `~/.aws/` , and the EC2/instance profile when applicable. Use `-p / --profile` to force a named profile from the shared config.

Inline keys: `--aws-access-key-id` and `--aws-secret-access-key` together install a static credential provider for the whole run. Add `--aws-session-token` when using STS temporary credentials. This mode is **mutually exclusive** with `--aws-credentials-file` . Inline secrets are visible in **process listings** and shell history; prefer env vars, a credentials file, or an IAM role when possible.

Credentials file: `--aws-credentials-file` points at an INI file in the same format as `~/.aws/credentials` (`[profile-name]` sections and `aws_access_key_id` / `aws_secret_access_key` / optional `aws_session_token`).

Without `-p`, the SDK uses the `default` profile from that file; with `-p myprofile`, the `[myprofile]` section is used.

Proxied Probing

When ELBaph executes active probing checks (such as `alb-external-reachability`), it performs HTTP connections directly to backend target IPs to determine if they are exposed to the public internet.

Note: Active probing is still under active development. Depending on network topology, proxy behavior, and target protocol specifics, results may be incomplete or inaccurate. Treat probe findings as strong signals that should be validated manually

If you are running ELBaph from within a corporate VPC or behind an enterprise VPN, your machine may possess privileged network access to these targets. This internal routing can result in ELBaph falsely classifying an isolated instance as "reachable from the internet".

To mitigate these false positives, use the `--proxy` flag to route the active probes through an external vantage point (e.g., an internet-facing proxy server). This ensures that reachability tests accurately reflect the perspective of an external, unauthenticated attacker.

For `alb-vhost-probe` on **HTTP** listeners, an `http://` proxy (including Burp on `127.0.0.1:8080`) is implemented with **CONNECT to alb-dns:port**, then a normal `GET` to `--vhost-probe-path` (default `/`) whose `Host` header is the vhost candidate. That avoids a Go `net/http` quirk where a plain proxied request would otherwise send `GET http://<Host>/...` and the proxy would contact the vhost name instead of the load balancer. Ensure your proxy allows **CONNECT to port 80** when testing HTTP listeners (Burp: *Proxy settings Request listeners allow ...* or add an upstream rule as needed). With more than **50** candidate hosts, vhost probes run **concurrently**; cap parallelism with `ELBAPH_VHOST_PROBE_MAX_CONCURRENCY` (default **8**).

Output

Every scan creates a timestamped folder in the current working directory:

```
elbaph-2026-04-01_22-00-00/
  summary.txt      run metadata + per-check finding counts
  albs.txt         table of all discovered ALBs
  nlbs.txt         table of all discovered NLBs (scheme, DNS, listeners)
  target_groups.txt table of all target groups + health
  rules.txt        per-ALB listener rule chain dump
  findings.txt     one detailed block per finding
  vhost_route53_attachments.txt Route 53 FQDNs ALB (alias/cname) for alb-vhost-probe
  vhost_route53_corpus.txt    record-owner FQDNs from listed zones, excluding zone apex (probe wordlist source
  topology.html    vis-network resource map (VPC ALB listener rule TG targets; multi-ALB targets hi
```

The `topology.html` file is an interactive diagram of how your ELBs sit in the VPC and how traffic is routed through listeners, rules, target groups, and backend targets.

Findings are also **printed live to the terminal** as each check completes - you do not have to wait for the full scan to finish.

If `--output` is specified, a `report.json` / `report.md` / `report.sarif` is written into the same folder.

Required AWS Permissions

ELBaph is **read-only**. The minimum IAM policy required is:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "elasticloadbalancing:DescribeLoadBalancers",
        "elasticloadbalancing:DescribeLoadBalancerAttributes",
        "elasticloadbalancing:DescribeListeners",
        "elasticloadbalancing:DescribeRules",
        "elasticloadbalancing:DescribeTargetGroups",
        "elasticloadbalancing:DescribeTargetHealth",
        "ec2:DescribeRegions",
        "ec2:DescribeInstances",
        "route53:ListHostedZones",
        "route53:ListResourceRecordSets"
      ],
      "Resource": "*"
    }
  ]
}
```

Optional Route 53 permissions enrich `alb-vhost-probe` (FQDN corpus and alias correlation).
Details: **[docs/elbaph-checks-documentation.md](https://docs.elbaph-checks-documentation.md)**.

Checks

Built-in check IDs (for `-c` / `--checks`). **Methodology, sample output, and how to read each finding:** **[docs/elbaph-checks-documentation.md](https://docs.elbaph-checks-documentation.md)**.

ID	Name
alb-client-ip-spoofable	Client IP Spoofing via Preserved X-Forwarded-For
cloudfront-bypass	CloudFront Origin Direct Access
alb-desync-monitor	Desync Mitigation Mode is Monitor
alb-external-reachability	Target External Reachability
alb-ip-gate-bypass	IP Gate Bypass via Alternate ALB
alb-optional-auth	Optional Authentication Allowed
alb-rule-shadowing	Restrictive Listener Rule Shadowed by Broader Earlier Rule
alb-vhost-probe	ALB virtual host routing probe
nlb-direct-target-reachability	NLB Target Directly Reachable

Adding a New Check

A check is a single Go file in `internal/checks/`. The only contract is the `Check` interface:

```
type Check interface {  
    ID()      string  
    Name()    string  
    Description() string  
    Run(ctx context.Context, data *collector.Dataset) ([]model.Finding, error)  
}
```

Step-by-step

1. Create the file

```
internal/checks/my_check.go
```

2. Implement the interface

```
package checks
```

```
import (
```

```
    "context"
```

```
    "fmt"
```

```
    "github.com/doyensec/elbaph/internal/collector"
```

```
    "github.com/doyensec/elbaph/internal/model"
```

```
)
```

```
/*
```

```
alb-my-check
```

Describe what this check looks for, why it matters, and what the exact trigger conditions are. This comment is the primary documentation for anyone reading or reviewing the check.

Trigger condition:

ALB scheme = internet-facing

...

```
*/
```

```
func init() { Register(&MyCheck{}) }
```

```
type MyCheck struct{}
```

```
func (c *MyCheck) ID() string {
```

```
    return "alb-my-check"
```

```
}
```

```
func (c *MyCheck) Name() string {
```

```
    return "alb-my-check"
```

```
}
```

```
func (c *MyCheck) Description() string {
```

```
    return "One-line description of what this check detects."
```

```
}
```

```
func (c *MyCheck) Run(ctx context.Context, data *collector.Dataset) ([]model.Finding, error) {
```

```
    var findings []model.Finding
```

```
    for _, lb := range data.LoadBalancers {
```

```
        // ... detection logic ...
```

```
        findings = append(findings, model.Finding{
```

```

ID:      fmt.Sprintf("%s | %s", c.ID(), lb.ARN),
CheckID:  c.ID(),
CheckName: c.Name(),
Region:   data.Region,
LoadBalancerARN: lb.ARN,
LoadBalancerName: lb.Name,
LoadBalancerScheme: lb.Scheme,
Detail:   `Explain what was found and why it is a problem.`,
Remediation: `Explain what should be done to fix it.`,
})
}

return findings, nil
}

```

3. That's it.

The `init()` call registers the check globally. No other file needs to be modified. The new check will appear automatically in `--checks list` output and run as part of the default scan. Add a short entry (ID, name, narrative, sample output) to **docs/elbaph-checks-documentation.md** and a one-line row in the README checks table.

What data is available

The `collector.Dataset` passed to `Run()` contains everything collected from AWS:

Field	Type	Contents
LoadBalancers	<code>[]model.LoadBalancer</code>	All ALBs with listeners, rules, wired target group pointers, and <code>Route53Names</code> (FQDNs that ALIAS/CNAME to the ALB when Route 53 APIs are permitted)
TargetGroups	<code>map[string]*model.TargetGroup</code>	All TGs keyed by ARN, with health data

Helper functions in `internal/albutil/` :

Function	Purpose
<code>albutil.ConditionSummary(conds)</code>	Compact string representation of rule conditions
<code>albutil.ActionChainDesc(actions)</code>	Ordered slice of human-readable action descriptions

The `pkg/netutil` package provides `IsRFC1918(ip)` .

Project Structure

```
elbaph/
  docs/
    elbaph-checks-documentation.md    detailed check documentation + sample findings output
  main.go
  cmd/
    root.go      cobra root command
    scan.go      scan subcommand - orchestrates collect  check  output
    version.go
  internal/
    model/       shared domain types (LoadBalancer, Finding, ScanMeta, ...)
    collector/   AWS API calls  model types (no SDK types leak out)
    checks/      detection modules + shared registry
    output/      always-on text folder + live terminal printer
    export/      optional structured exporters (JSON, Markdown, SARIF)
  pkg/
    netutil/     RFC1918 and CIDR helpers
```

:handshake: Contributing

ELBaph thrives on community contributions. Whether you are a developer, researcher, or bug hunter, your feedback and pull requests help improve the project for everyone. Please use the [Git Hub issue tracker](#) for reports and discussion.

This project is developed with support from [Doyensec](#).

[\[image: Doyensec Research\]](#)

docs/elbaph-checks-documentation.md

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ELBaph checks reference

This document describes each built-in check: what it looks for, how it works, and how to read sample terminal / `findings.txt` output. Check IDs match the `-c` / `--checks` flag.

alb-client-ip-spoofable

Name: Client IP Spoofing via Preserved X-Forwarded-For

What it detects: Internet-facing ALBs with `routing.http.xff_header_processing.mode` set to `preserve` , so a client-supplied `X-Forwarded-For` is passed through unchanged. Backends that trust that header for access control or auditing can be fooled.

How it works: Passive - reads ALB attributes from the collector; no HTTP probe.

Sample output

Check: alb-client-ip-spoofable - Client IP Spoofing via Preserved X-Forwarded-For

Region: us-east-1

ALB: my-alb (internet-facing)

Detail:

The internet-facing ALB is configured to "preserve" the X-Forwarded-For header.

In this mode, the load balancer passes any client-supplied X-Forwarded-For header directly to the target group without modification. [...]

PoC:

Spoof internal IP 10.0.0.1 via X-Forwarded-For header:

```
curl -H "X-Forwarded-For: 10.0.0.1" http://my-alb-123.us-east-1.elb.amazonaws.com/
```

You get one finding per affected ALB. The Detail block explains the attribute mode; PoC shows how to inject a fake client IP.

cloudfront-bypass

Name: CloudFront Origin Direct Access

What it detects: An internet-facing ALB or NLB is registered as a CloudFront origin, but the origin is still reachable directly (HTTP/HTTPS probe succeeds), so edge controls (WAF, geo, etc.) can be skipped.

How it works: Uses CloudFront to origin mapping from the collector, then active probes to the load balancer DNS (optionally with Host / SNI hints from listener rules or cert SANs).

Sample output

Check: cloudfront-bypass - CloudFront Origin Direct Access

ALB: api-alb

Detail:

CloudFront is configured as a distribution in front of [this](#) load balancer, but the origin is directly reachable [from](#) the internet. [...]

Resource: ALB CloudFront distribution: d111111abcdef8.cloudfront.net

Verdict: origin reachable with Host=api.example.com via HTTPS [200](#)

PoC:

curl -vk -H 'Host: [api.example.com](#)' https://api-alb-123.us-east-1.elb.amazonaws.com:443/

Verdict says whether the origin answered and with which protocol and status. Resource tells you ALB vs NLB. You may see several findings when multiple host headers are probed.

alb-desync-monitor

Name: Desync Mitigation Mode is Monitor

What it detects: ALB `routing.http.desync_mitigation_mode` is `monitor`, so non-RFC7230 requests are forwarded to targets instead of being blocked.

How it works: Passive - attribute from DescribeLoadBalancerAttributes.

Sample output

Check: alb-desync-monitor - Desync Mitigation Mode is Monitor

ALB: legacy-alb

Detail:

The ALB's HTTP desync mitigation mode is set to `"monitor"`.

In [this](#) mode, the load balancer passes incoming requests that [do](#) not comply with RFC [7230](#) directly to the target group without blocking or modifying them. [...]

Configuration-only; there is no PoC. Remediation is to switch to `defensive` or `strictest`.

alb-external-reachability

Name: Target External Reachability

What it detects: A registered target has a public IP and responds to HTTP/HTTPS on the backend port from the scanner's vantage point, bypassing ALB rules and WAF.

How it works: Active probes to `publicIP:port` (respects `--proxy`).

Sample output

Check: alb-external-reachability - Target External Reachability
ALB: web-alb
Listener: HTTP:80
Rule: priority 10
Target: i-0abc123 (10.0.0.0 is example; real output shows public IP)

Detail:

Target "i-0abc123" with public IP "203.0.113.50" is directly reachable from the scanner's vantage point over HTTP (returned HTTP 200). This bypasses any ALB-level protection such as WAF, authentication actions, or listener path routing.

PoC:

```
curl -v http://203.0.113.50:8080/
```

Target and public IP identify the bypass path. Listener and rule show which ALB path would have applied if traffic went through the ALB.

alb-ip-gate-bypass

Name: IP Gate Bypass via Alternate ALB

What it detects: Targets sit behind a source-ip-gated rule on one public ALB, but the same targets are also reachable through another public ALB or listener path with no source-ip condition.

How it works: Passive cross-reference of target IDs across rules and ALBs.

Sample output

Check: alb-ip-gate-bypass - IP Gate Bypass via Alternate ALB

ALB: gated-alb

Detail:

A source-ip condition on one ALB creates the appearance of IP-gated access [...]

IP-gated path:

ALB "gated-alb" listener HTTPS:443 priority 10 condition: source-ip 203.0.113.0/24

Target group: "tg-app" (type instance)

Targets registered in gated TG (2): i-aaa, i-bbb

Bypass applies to these targets (also on an ungated path) (2): i-aaa, i-bbb

Alternate (ungated) paths (grouped by route):

- "open-alb" / TG "tg-app" HTTPS:443 pri 20 path /api* | open-alb-123.elb.amazonaws.com

bypassed: i-aaa, i-bbb

TG instance (2): i-aaa, i-bbb

PoC:

Query the ungated ALB instead of the IP-gated one.

curl -vk https://open-alb-123.elb.amazonaws.com:443/

The bullet list under alternate paths is the bypass: same instance IDs with a weaker rule. Run PoC against the ungated ALB DNS.

alb-optional-auth

Name: Optional Authentication Allowed

What it detects: A rule contains `authenticate-cognito` or `authenticate-oidc` with

`OnUnauthenticatedRequest=allow`, so unauthenticated clients are forwarded to the backend.

How it works: Passive inspection of listener rule actions.

Sample output

Check: alb-optional-auth - Optional Authentication Allowed

Listener: HTTPS:443

Rule: priority 5

Condition: path /admin*

Actions: authenticate-oidc, forward -> tg-admin

Detail:

The authenticate-oidc action has OnUnauthenticatedRequest set to "allow".

Unauthenticated users bypass the IdP entirely and are forwarded directly to the backend. [...]

PoC:

`curl -vk https://alb.../ # send request without auth cookies/headers - expect 200`

Condition and actions show which path looks protected but is not.

alb-rule-shadowing

Name: Restrictive Listener Rule Shadowed by Broader Earlier Rule

What it detects: A later rule applies stronger controls (auth, JWT-style headers, source-ip allowlist heuristics), but an earlier rule matches a superset of traffic and forwards without those controls, so the restrictive rule never runs for that traffic.

How it works: Passive ordering and condition heuristics on public ALBs.

Sample output

Check: alb-rule-shadowing - Restrictive Listener Rule Shadowed by Broader Earlier Rule

Listener: HTTPS:443

Detail:

Rule priority 10 (path-pattern /*) is a superset of rule priority 20 (host-header api.example.com + authenticate-oidc).

Because rules are evaluated in ascending priority order, traffic intended for the restrictive rule 20 will match the broader earlier rule 10 instead, bypassing that rule's controls [...]

A lower priority number is evaluated first. Fix by reordering or narrowing the broad rule.

alb-vhost-probe

Name: ALB virtual host routing probe

What it detects: For an internet-facing HTTP or HTTPS listener, response fingerprints differ between (1) a random baseline `Host` and (2) candidate names from listener host-header rules, Route 53 names that alias or CNAME to the ALB, a Route 53 corpus (all record owners in listed zones except zone apex), and an optional user-supplied host list file. A difference suggests a distinct routing path for that `Host`.

How it works: Active `GET` to `http(s)://<ALB-DNS>:<port><path>` with `Host` set per candidate. The path comes from `--vhost-probe-path` (default `/`). TLS SNI uses the ALB hostname. With an `http://` proxy, plain-HTTP listeners use `CONNECT` to `alb:port` then HTTP on the tunnel (see README proxy section). Candidate cap: `ELBAPH_VHOST_PROBE_MAX_CANDIDATES` (default 5000). If there are **more than 50** candidates, candidate probes run **concurrently**; limit workers with `ELBAPH_VHOST_PROBE_MAX_CONCURRENCY` (default 8, max 64).

Scan flag	Meaning
<code>--vhost-probe-hosts-file <path></code>	Newline-separated hostnames merged after listener host-header values and Route 53 alias/CNAME names for this ALB, but before the Route 53 zone corpus . That order avoids losing your file when the corpus alone would fill <code>ELBAPH_VHOST_PROBE_MAX_CANDIDATES</code> . Lines starting with <code>#</code> and blank lines are ignored; duplicates within the file are dropped (first-seen order). Findings label these as Source: file-corpus . The global merge still dedupes across tiers (earlier tier wins for the same host).
<code>--vhost-probe-path <path></code>	HTTP path for baseline and every candidate probe (e.g. <code>/</code> or <code>/status</code>). Must start with <code>/</code> ; normalized on the scan command if you omit a leading slash. Use a path that exists on your apps if you care about comparable responses; the default <code>/</code> may not match a real route on every stack.

Fingerprints: HTTP status, full body SHA256, normalized Location, Content-Length, sorted Set-Cookie names (values not compared). The finding prints a baseline request/response block, then each candidate with Changed fields and a full candidate fingerprint block.

Sample output (finding detail)

Listener: HTTPS:443

Baseline request (randomized Host header):

Request to: https://k8s-ui-123.eu-west-1.elb.amazonaws.com:443/

Method: GET

Path: /

Host header: invalid.elbaph-baseline-a1b2c3d4.invalid

Baseline response (fingerprint):

Status: HTTP/1.1 200 OK

Body SHA256: 19ba3f1f7a7d58bc7b72c3639d8b9c89c4ce569d434da7964945e5ab657ed028

--- Candidate 1 of 2 ---

Host header **for this** probe: api.example.com

Source: route53-alias

Changed fields: status, body, content-length

Candidate fingerprint:

HTTP status: 404

Body SHA256: 99eb12f2ab3c4866a353e098ffa3cb7a967e617c49b98480394ec5d8ea92b094

Location: (none)

Content-Length: 18

Set-Cookie names: (none)

PoC:

Baseline (TCP/TLS to ALB only; Host header is the random baseline name)

curl -vk -H 'Host: elbaph-baseline-a1b2c3d4.invalid' https://k8s-ui-123.eu-west-1.elb.amazonaws.com:443/

Candidate api.example.com (same URL as baseline; only Host changes)

curl -vk -H 'Host: api.example.com' https://k8s-ui-123.eu-west-1.elb.amazonaws.com:443/

Baseline request shows the exact ALB URL, path, and baseline Host. Baseline response is the default-rule fingerprint. Each candidate section lists which fingerprint fields differ (Changed fields) and the full candidate fingerprint. This is not proof of vulnerability by itself. Artifacts:

vhost_route53_corpus.txt , vhost_route53_attachments.txt .

nlb-direct-target-reachability

Name: NLB Target Directly Reachable

What it detects: An NLB target has a public address and accepts TCP or TLS on the registered backend port from the scanner, bypassing the NLB.

How it works: Active probe (ProbeNLBBackend) with optional SNI from NLB DNS.

Sample output

Check: nlb-direct-target-reachability - NLB Target Directly Reachable

NLB: nlb-api

Listener: TLS:443

Detail:

NLB "nlb-api" forwards TLS:443 to instance target i-0abc with public IP 203.0.113.50.

Active probe to 203.0.113.50:8443 succeeded (TLS handshake OK). Direct access bypasses the NLB.

PoC:

openssl s_client -connect 203.0.113.50:8443 -servername "nlb-123.elb.amazonaws.com" -brief

The port in the detail is the target or backend port, not necessarily the listener front port. Restrict security groups so only the NLB can reach that port.

Environment and IAM notes

Concern	Notes
--proxy	Reachability and many probes honor it. For alb-vhost-probe on HTTP listeners with an http:// proxy, traffic uses CONNECT to the ALB; allow CONNECT to port 80 on the proxy if needed.
Route 53	alb-vhost-probe uses ListHostedZones and ListResourceRecordSets when permitted; without them, vhost candidates fall back to listener host-header values, the optional --vhost-probe-hosts-file list, and alias names already on the ALB.

See the main README for installation, flags, and IAM JSON.

Archived from <https://github.com/doyensec/elbaph>. Rendered offline from the local Markdown copy.